Review

The Chain Rule states that:

$$\frac{d}{dx}\left[f(g(x))\right] = _____$$

By taking the antiderivatives of both sides, we get:

$$\int _____ \, dx = _____$$

Method of u-substitution

Suppose we want to evaluate $\int f'(g(x))g'(x)\,dx$. Let $u = g(x)$.

Then $du =$ _____ dx.

Hence,

$$\begin{aligned}\int f'(g(x))g'(x)\,dx &= _____ \\ &= _____ \\ &= _____\end{aligned}$$

Note also that

$$\begin{aligned}\int_a^b f'(g(x))g'(x)\,dx &= _____ \\ &= _____\end{aligned}$$

Examples

$\int x(1+x^2)^5 \, dx$

$\int_0^{\frac{\pi}{2}} \sin x \cos x \, dx$

$\int e^{kx} \, dx$ (Note: we already did this by a guess-and-check method on the previous sheet - let's do it in a more rigorous way.)

Questions

Compute each of the following integrals. For the indefinite integrals, check your answer by differentiating (this is part of the graded work!). Integrals done using u-substitution are most often checked using the chain rule.

Note: Not all of these require substitutions. One of the most important integration skills is spotting what is the easiest method to use for a given integral.

1. $\int_0^1 \frac{x}{\sqrt{x^2+1}}\,dx$

□Regrade

2. $\int \frac{1}{3x+1}\,dx$

□Regrade

3. $\int \frac{1}{e^{3x}}\,dx$

□Regrade

4. $\int \sin^2 x\,dx$ (Hint: $\cos 2x = 1 - 2\sin^2 x$. Rearrange this equation, then use your work to rewrite this integral.)

□Regrade

5. $\int_0^2 \frac{e^x}{1+e^{2x}}\,dx$ (Hint: What's another way to write e^{2x}? Rules of exponentials...)
□Regrade

6. $\int \frac{\ln x}{x}\,dx$
□Regrade

7. $\int \cos^2 x \sin x \, dx$

☐Regrade

8. $\int x e^{-x^2} \, dx$

☐Regrade

9. $\int \tan x \, dx$ (Hint: write $\tan x$ as a quotient of two other trig functions.)
□Regrade

10. $\int \frac{2x}{\sqrt{1-x^4}} \, dx$
□Regrade

11. $\int_0^1 x\sqrt{x+1}\,dx$ (Hint: Let $u = x + 1$. Then $x =$...?)

□Regrade